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Somatic Symptoms of Anxiety: Comparison of Self-Report and Physiological Measures

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The frequently reported absence of significant correlations between patient rating scales and physiological measures has led to the belief that patients cannot reliably perceive physiological changes that are experienced under conditions of stress. To determine whether or not this conclusion is justified for patients with clinical anxiety, self-reports and psychophysiological recordings were examined and compared in 20 patients suffering from generalized anxiety disorder. No systematic correlations were found between patient ratings and physiological measures of somatic symptomatology during periods of rest or psychological stress (Stroop Test). However, parallel directional changes in the two sets of measures were observed upon exposure to stress, indicating that patients could accurately report the direction, but not the degree, of changes in physical symptoms of anxiety. These results suggest that patient reports of physical symptoms such as sweating and rapid heart rate can be useful in clinical evaluation and research settings that do not require quantitative assessment of physiological activity.

Introduction

In anxiety disorders, the selection of treatment and its assessment depends, to a large extent, on the level of somatic symptomatology (Hoehn-Saric and McLeod 1985). It is important, therefore, to determine how well subjective complaints, upon which most treatments and studies rely, correspond to actual physiological activity. For example, in assessing a person's complaint of muscular tension, we would like to know the extent to which the symptoms correspond to the actual electromyographic (EMG) activity.

Several studies have examined the relationships between self-reports and physiological processes (Tyrer et al. 1980; Pennebaker 1981; Katkin et al. 1982; Pennebaker et al. 1982) and, although significant correlations occasionally are found (e.g., Kelly and Walter 1968), there is a general agreement that subjective report and physiology do not correlate very well (Pennebaker 1982; Ray et al. 1983). Low correlations between self-reports and physiological activity do not necessarily mean, however, that patients cannot perceive physiological changes related to stress. They may simply reflect a quantitative inaccuracy, rather than an inability to perceive changes. For example, if the correlation between self-report of sweating and measured skin conductance is low for a group of patients, this

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may mean only that those patients who report high levels of perspiration are not necessarily the ones who perspire the most. It is still possible that increases in skin conductance are accompanied by increases in patients' self-reports of sweating. If this is so, if self-report and physiological activity move in the same direction under stress, then patients can accurately perceive the *direction* of physiological change. In this case, a low correlation between self-report and physiological state would simply reflect a perceptual inaccuracy in terms of the *degree* of change.

In the following study, we provide a comparison of self-report and physiological measures under conditions of rest and stress in a clinically anxious population. We then discuss our findings as they relate to the reliability of patient self-reports.

Methods

Subjects

Subjects consisted of 20 patients who were diagnosed, following guidelines similar to those of the Structured Clinical Interview for DSM-III (SCID) (Spitzer and Williams 1984a,b), as having generalized anxiety disorder, as specified in DSM-III (American Psychiatric Association 1980). In addition, participants were required to meet the following criteria: a score of 18 or above on the Hamilton Anxiety Scale (Hamilton 1959), a score of 8 or above on the Covi Anxiety Scale (Covi et al. 1979), and a score of 38 or above on the Trait form of the State–Trait Anxiety Inventory (Spielberger et al. 1970). The means and ranges of the rating scales are presented in Table 1. Excluded were patients with major mental illness, history of psychosis, or history of recent substance abuse. Subjects had to be sufficiently literate to fill out questionnaires, and they had to be free of medications affecting the central and autonomic nervous systems for at least 1 week. Of the 20 patients, there were 10 men and 10 women, 18 were white, 1 was black, and 1 was an American Indian. The average age was 39.2 years (range 23–60), and the average educational level was 15.0 years (range 9–28). Following an explanation of the study procedures, participants gave their informed consent in writing.

Patients participated in an experimental session that consisted of the completion of the investigator and patient rating scales and psychophysiological measurements described below.

Table 1. Anxiety Rating Scales (N = 20)

Scale	Mean score ± SD
Trait	54.4 ± 10.4
Covi	10.7 ± 1.8
State	51.4 ± 8.0
HAM (T)	28.0 ± 4.8
HAM (CV)	1.9 ± 1.0
HAM (M)	2.2 ± 0.9
HAM (AUT)	2.2 ± 0.7
SSS (T)	21.2 ± 9.4
SSS (HR)	1.2 ± 1.0
SSS (M)	4.0 ± 2.5
SSS (SW)	2.2 ± 1.8

SSS, Somatic Symptoms Scale; HAM, Hamilton Anxiety Scale; T, total; HR, heart rate; M, muscular; SW, sweating; CV, cardiovascular; AUT, autonomic

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Therapist Ratings

Hamilton Scale of Anxiety (HAM: Hamilton 1959). This is a 14-item interviewer-rated scale that clearly differentiates between somatic and psychic symptoms of anxiety. Patients were rated at the beginning of the session with focus on their level of functioning during the preceding week, as well as on the morning of testing. This scale provided an observer's estimate of a patient's level of somatic symptoms, as well as a criterion score for inclusion of patients in the study.

Covi Anxiety Scale (Covi et al. 1979). This scale requires an interviewer to rate the patient's verbal report, behavior, and somatic symptoms of anxiety during the interview on a 15-point scale. A score of 8 or more is the accepted criterion for classifying patients as anxious when conducting research.

Patient Ratings

Somatic Symptoms Scale (SSS; Hoehn-Saric 1981). This scale consists of 21 items describing somatic symptoms that frequently accompany anxiety. Each item is rated on a 4-point severity subscale. Subscales were obtained by selecting items relating to sweating, heart rate, and muscular tension. This scale was completed before the session and provided a patient's estimate of somatic symptoms of anxiety over the preceding week, including the morning of testing.

State–Trait Anxiety Inventory (Spielberger et al. 1970). This questionnaire consists of two subscales, a Trait and a State scale, each containing 20 items. Both scales measure predominantly the psychic aspects of anxiety. The Trait form rates a person's general tendency to respond to events with anxiety. The State form describes the level of psychic anxiety at the time it is being administered. Patients completed both scales prior to the experimental session.

Assessment of Somatic Symptoms during Experimental Sessions

Body Feeling Analog Scale. This 9-item scale was adapted from that developed by Tyrer (1976) for assessing effects of anxiolytic drugs on somatic symptoms. Patients were asked to rate the way they felt at a given moment with respect to each of the items on an analog scale from "absent" to "very severe." Four of these items were used for the present study: "sweating," "shaking/trembling," "palpitations," and "muscular tension." These analog scales were completed during experimental sessions immediately following periods of rest and stress. Whereas the Hamilton Scale of Anxiety and the Somatic Symptoms Scale provided patients' ratings of physical symptoms over the preceding week, the analog scales provided patients' reports of physical symptoms at discrete times during the testing session.

Physiological Measures

The following measures were recorded by means of a Cyborg Bio-Lab (Stoelting Co., Chicago, IL) and an Apple II microcomputer with appropriate interfacing: heart interbeat interval (IBI), skin conductance (SC), and frontalis and gastrocnemius electromyographic

(EMG) activity. Blood pressure was obtained by using an automated blood pressure monitor (Infrasonde, Puritan Bennett Co.). Heart interbeat interval was obtained by placing disposable electrodes on standard sites for obtaining the electrocardiogram. Skin conductance was recorded by placing electrodes (silver/silver chloride, 1 cm diameter) on the volar surfaces of the left index and middle fingers (middle segments) and was monitored by means of a J&J Co. Electrodermograph. Frontalis and gastrocnemius EMGs were obtained by placing sets of three electrodes (silver/silver chloride, 15 mm diameter) over the respective muscle groups. All signals were amplified and stored in the microcomputer for later decoding and averaging; data were sampled every 200 msec and stored on disk at the end of each test segment. For heart interbeat interval analysis, outliers were defined as samples that deviated from immediately preceding sample values by more than 20%. Outliers were replaced by setting them equal to the immediately preceding value.

Stress Condition

Stroop Test (Stroop 1935). The Stroop Test induces stress by creating perceptual interference and has been shown to produce significant changes in psychophysiological measures (Newlin and Levenson 1982; Forsman and Lindblad 1983). To avoid the substantial effects of speech on physiological measures such as blood pressure and heart rate (Lynch et al. 1980; Friedmann et al. 1982), we decided to use an automated version of the test. Programmed for use with the Apple II microcomputer and a numeric key pad (The Keyboard Co.), the test consists of three parts. In the first part, the words "red," "green," and "blue" are randomly chosen and printed in white block letters in a column along the left half of the video screen. An arrow moves downward from one word to the next as the subject responds by pressing one of three keys on the numeric key pad, each of which corresponds to a different color: red, green, or blue. Following presentation of 100 words, the next part of the Stroop Test begins. In this part of the task, blocks with the colors red, green, or blue appear, rather than words, on the left side of the screen. The procedure for the subject is the same: press the key that corresponds to the color opposite the arrow. After presentation of 100 color blocks, the third and last part of the Stroop Test begins. In this part of the test, three groups of 100 words each are presented. Each word, "red," "green," or "blue" is presented in block letters colored in red, green or blue. Thus, for example, the word "red" might be printed in blue letters, or the word "blue" might be printed in green letters, etc. The subject must press the key associated with the color of the letters, not with the color spelled out.

Testing Procedure

Upon arriving on the morning of the test session, subjects were assessed by an interviewer on the Hamilton and Covi Anxiety Scales and completed the State–Trait Anxiety Inventory and the Somatic Symptoms Scale. Patients next sat in a comfortable reclining chair in an air-conditioned room kept in constant dim illumination. The experimental room contained an Apple II microcomputer with monitor (video screen) and numeric key pad placed within convenient reach of the patient. Electrodes were attached to the patient, as described above.

The session consisted of a series of psychophysiological assessments,[1] including a baseline rest period (5 min) and exposure to a stressful mental task (computerized version of the Stroop Test described above, 10 min). Stroop Test scores will be reported elsewhere. Patients rated themselves on the analog scales described above immediately following the baseline period and the Stroop Test.

Data Analysis

Correlations among subjective and physiological measures were examined. Pearson product–moment correlation coefficients were obtained for correlations on absolute levels of these measures, as well as on changes in measures from baseline to stress conditions. Comparisons of measures between baseline and stress conditions were made by means of two-tailed paired Student's *t*-tests. For these comparisons, the 5-min baseline segments were compared to the first 5 min of the Stroop Test. Variance measures of skin conductance and heart interbeat interval were estimated by means of the Mean Square Successive Difference statistic (Heslegrave et al. 1979).

Results

Comparison of the Effects of Stress on Patient Ratings and Physiological Measures

The effects of stress, both on analog ratings of somatic symptoms and on physiological recordings, are presented in Figure 1 and in Table 2. In Figure 1, patients' estimates of "sweating" (SW), "palpitations" (PAL), "muscle tension" (MUS), and "shaking/trembling" (S/T), as obtained from analog rating scales, are shown in the top panel, actual physiological measures of skin conductance (SC), heart interbeat interval (IBI), frontalis EMG (FEMG), and gastrocnemius EMG (GEMG) are presented in the bottom panel.

Patient reports and physiological measures of cardiac and eccrine sweat gland activity showed parallel changes upon exposure to stress. The *t*-tests revealed significant increases in analog "sweating," as well as in mean skin conductance. Also, increases in analog "palpitations" accompanied decreases in heart interbeat interval (increases in heart rate). Although gastrocnemius EMG activity increased under stress, no significant changes were seen in frontalis EMG activity or in analog ratings of "shaking/trembling" or "muscle tension."

Table 2 shows that, in addition to the changes seen in Figure 1, significant increases in spontaneous fluctuations in skin conductance, skin conductance variance, and skin conductance range occurred under stress, along with increases in systolic blood pressure.

Relationships between Patient Ratings and Physiological Measures

Correlations between analog ratings of somatic symptoms and the physiological measures revealed that ratings of "palpitations" were weakly related to average IBI ($r = -0.5$;

[1]Each session consisted of a baseline rest period (5 min), startle and habituation to tones (15 min), hyperventilation segment (30 sec hyperventilation plus 8 min recovery), and exposure to a stressful mental task (computerized version of the Stroop Test, 10 min). Total recorded time for each session thus was approximately 40 min.

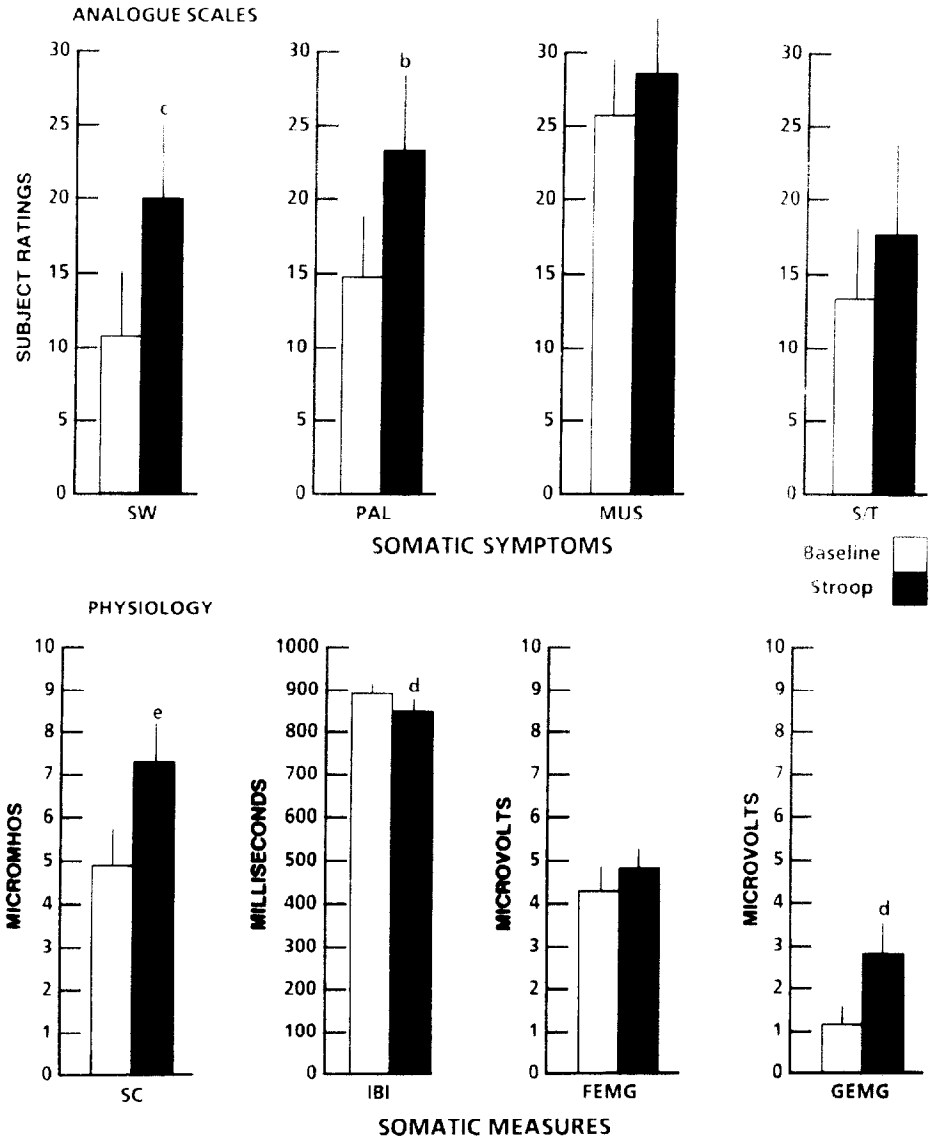

Figure 1. Effects of stress on somatic symptoms of anxiety. (Top) Analog rating scales. (Bottom) Physiological measures. X axis: Somatic symptoms and measures. Y axis: Subject ratings and physiological recordings. Lines above the histograms represent standard errors of the means (SEM) Abbreviations: SW, sweating; PAL, palpitations; MUS, muscular tension; S/T, shaking/trembling; SC, mean skin conductance level; IBI, mean heart interbeat interval; FEMG, frontalis EMG; GEMG, gastrocnemius EMG. Significance levels: b,$p < 0.05$; c,$p < 0.02$; d,$p < 0.01$; e,$p < 0.001$.

$p < 0.05$) and systolic blood pressure ($r = 0.44$; $p < 0.10$) during baseline and to IBI range ($r = 0.49$; $p < 0.05$) during the Stroop Test. No correlations between "sweating" and skin conductance, or between either "shaking/trembling" or "muscle tension" and EMG activity were found. Nor were correlations between change scores (Stroop − baseline) in these measures found.

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Table 2. Physiological Effects of Psychological Stress (N = 20)

	Baseline	Stroop
SC fluctuations (number)	2.8 ± 1.0	17.2 ± 5.4[c]
SC variance ($\times 10^3$ μmhos)	0.6 ± 0.1	2.4 ± 1.0[a]
SC range (μmhos)	1.3 ± 0.3	2.8 ± 0.4[d]
IBI variance (msec)	873.2 ± 151.8	753.8 ± 128.4
IBI range (msec)	255.1 ± 15.4	316.5 ± 45.4
Systolic BP (mm Hg)	113.8 ± 13.6	118.2 ± 14.2[b]
Diastolic BP (mm Hg)	73.7 ± 9.3	75.0 ± 7.6

SC, skin conductance; IBI, heart interbeat interval; BP, blood pressure. Data are presented as session means; standard deviations are presented for blood pressure and standard errors for all other measures.

[a] $p < 0.10$.
[b] $p < 0.02$.
[c] $p < 0.01$.
[d] $p < 0.001$.

Correlations between physiological recordings and structured interview scales were also absent, except for a positive correlation between the Somatic Symptom Scale (SSS) scores and diastolic blood pressure during the Stroop segment of the session (r = 0.54; $p < 0.02$).

Correlations among Rating Scales

Mean scores and standard deviations for the structured rating scales used are presented in Table 1. These scores were within the ranges typically found with clinically anxious patients. Correlations between investigator-rated and patient-rated scales are presented in Table 3. Investigator ratings of cardiovascular symptomatology and total symptomatology

Table 3. Rating Scale Correlations (N = 20)

Scales	r
State vs. HAM (T)	0.16
State vs. SSS (T)	0.49[b]
HAM (T) vs. SSS (T)	0.61[d]
HAM (CV) vs. SSS (HR)	0.78[d]
HAM (M) vs. SSS (M)	0.12
HAM (AUT) vs. SSS (SW)	−0.11
Trait vs. Covi	0.66[c]
Trait vs. State	0.27
Trait vs. HAM (T)	0.63[c]
Trait vs. SSS (T)	0.60[c]
Covi vs. State	0.20
Covi vs. HAM (T)	0.42[a]
Covi vs. SSS (T)	0.52[c]

See Table 1 for abbreviations.

[a] $p < 0.10$.
[b] $p < 0.05$.
[c] $p < 0.01$.
[d] $p < 0.001$.

correlated well with patient ratings. In addition, patients' ratings on the Trait form of the State–Trait Anxiety Inventory correlated highly with SSS total score, HAM total score, and Covi score. The State scores, on the other hand, correlated only with the SSS total score.

Discussion

Ray et al. (1983) have suggested that assessment procedures could benefit from a better understanding of how psychophysiological and psychological processes are related. In the present study, we examined such relationships in a generalized anxiety disorder population under conditions of rest and stress induced by an automated version of the Stroop Test.

Upon exposure to stress, parallel increases in analog ratings of "sweating" and measures of actual skin conductance were observed, along with parallel increases in analog ratings of "palpitations" and measured heart rate. These findings are consistent with those of Knight and Borden (1979), who measured patient ratings of "nervousness" rather than somatic symptoms. In socially anxious subjects exposed to social stress, parallel increases were found in "nervousness" and measures of skin conductance and heart rate. On the other hand, estimates of muscle activity and measured EMG activity did not change significantly in the same direction. That is, changes in some, but not all, measures were seen under stress. As Lacey (1959) has pointed out, one cannot generalize from one set of psychophysiological measures to another. In summary, patient reports of sweating and heart rate appear to be reliable directional estimates of changes in skin conductance and heart rate. Self-reports of muscle tension, however, are not reliably reflected by the EMG measures used in the present study (frontalis and gastrocnemius muscles).

Significant changes were seen in both analog and physiological measures during stress. Systematic correlations between patient ratings and physiological measures, however, were not. Similar results have been found in studies using other subject populations. For example, Natelson et al. (1983) found no significant correlations between subjective and physiological measures in healthy men during bedrest, and Morrow and Labrum (1978), examining victims of a first myocardial infarction, also failed to find significant correlations (see also, Pennebaker 1982; Ray et al. 1983). In the present study, however, we did find significant correlations between analog ratings of "palpitations" and measures of cardiovascular activity. These findings are consistent with those of Kelley et al. (1970) and Pennebaker (1982). It appears that perception of heart beat is quantitatively more accurate than that of other physiological processes involved in anxiety.

Past studies (e.g., Morrow and Labrum 1978; Winokur et al. 1982) have reported significant correlations among structured rating scales. This was also the case in the present study. Significant correlations among the Trait form of the State–Trait Anxiety Inventory, the Covi Anxiety Scale, the Hamilton Anxiety Scale, and the Somatic Symptoms Scale were found, providing evidence of construct validity. In addition, the fact that the SSS correlated with all the other anxiety scales highlights the prevalence of somatic symptoms in generalized anxiety disorder.

In conclusion, although the literature suggests that there is little correspondence between patient reports of physiological activity (i.e., symptoms) and objectively obtained physiological measures, the data from the present study show that patient reports and physiological measures can change in the same direction under stress. Thus, patient reports

of the existence of at least some physical symptoms (e.g., sweating, rapid heart rate) are reliable. The fact that subjective ratings correlated poorly, if at all, with the physiological measures suggests, however, that patient reports about the extent of their physical symptoms are not reliable. Patients showing high levels of actual physiological activity are not necessarily those whose ratings of physical symptoms are high, and vice versa. Thus, when accurate quantitative assessment of somatic symptoms is not required, self-reports of somatic symptoms can be useful both in clinical evaluation and in research. When accurate quantitative assessment of somatic change is required, however, one cannot rely on patient reports of physiological activity, but must obtain measures of the physiological activity itself.

In the present study, patients' responses to only one type of stressor were assessed. It is possible that different results would have been obtained under different conditions of stress. For example, responses to situations involving risk-taking may differ from those seen under conditions of cognitive stress. Different responses also may occur to situations more closely related to patients' individual psychopathologies. Future studies should examine the relationship between patient reports and physiological measures under a variety of conditions.

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